

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0610

Red Hill 345 kV
August 23, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0610. The Interconnection Customer proposes a 377.1 MW natural gas combined cycle facility interconnecting at the Red Hill 345 kV substation in the Virginia area. The request was received on April 18, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 273 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,982,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Grove Street Substation - Charlottesville 138 kV L	Red Hill - Fluvanna P.S 345 kV	115.2	240	21.9
Bluff City Substation - Reach 138 kV	Red Hill - Fluvanna P.S 345 kV	110.9	442	17.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Charlottesville - Grove Street 138 kV	Rebuild line with higher-capacity conductor	\$15,021,000	28% (\$4,191,000)
Bluff City - Reach Substation 138 kV L	Rebuild line with higher-capacity conductor	\$24,593,000	15% (\$3,791,000)
	Total allocated to project		\$7,982,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.